

Heat in Medicine

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Heat Therapy in Medicine: Techniques and Applications

Objective

Students will understand the various methods of heat therapy in medicine, their mechanisms, applications, and the physiological effects of heat on the human body. By the end of the lesson, students will be able to identify different heat therapy techniques and discuss their appropriate uses in clinical settings.

Recommended text book:

- 1. Medical Physics for Medical students (Taki AM Almusa) 2024**
- 2. Medical physics (John R. Cameron) 1978 ,1993,1999,2003 and 2010**
- 3. Introduction of Health Physics (Herman Cember) 2009**
- 4. Introduction to physics in modern medicine (Suzanne A. kane) 2010**
- 5. Electronics in medicine and biomedical instrumentation (Nandini K. Jog) 2006**
- 6. Radiation protection and dosimetry (Michel G. Stabin) 2012**
- 7. Dr.R.N.Roy .Atext book of Biophysics 1st edition,2005**
- 8- Physics of radiotherapy Khan 2012**
- 9- Advance in Medical Physics, Devon J. Godfrey, Volume 6 2016**

Heat therapy

Heat was recognized as therapeutic agent several thousand years ago. It has two primary therapeutic effects.

1. An increase in blood supply
2. An increase in metabolism resulting in relaxation of the blood capillaries, and then to cool down the heated area.

Heat production for therapy

1 – The conductive method:

Heat can transfer by conduction, the quantity of heat transfer depends on the temperature difference, the time of contact, the area of contact, and the thermal conductivity of the materials.

This can be done by several ways such as hot bath, hot packs and electric heating pad. This can lead to local surface heating .

2- Radiant Heat:

Heat radiation can be achieved by using infrared (IR) radiation. It penetrates about 3mm in the skin. It can be produce by glowing coils and by 250Watts incandescent lamps. The wavelengths used are between (800-4000nm). An excessive exposure can cause erythematic, sometimes swelling (edema) longer exposure cane cause skin browning or hardening. It is considered to be more effective than conductive heating because it can penetrate deeper.



3- Diathermy:

Short wave diathermy: - utilized electromagnetic wave in radio range (~10m) and microwave range (~12cm) short wave diathermy penetrate deep into tissue (more than conductive and radiant).

Different ways in transferring heat into the body in short wave diathermy.

A- The part of the body to be treated is placed between two plates (electrodes) connected with high frequency power supply. The charged particles of the tissue will be attracted to one plate and to the other depending upon the sign of the alternating voltage on the plate. This movement will produce heat (resistive-Joule) heating. Fig (4.9).

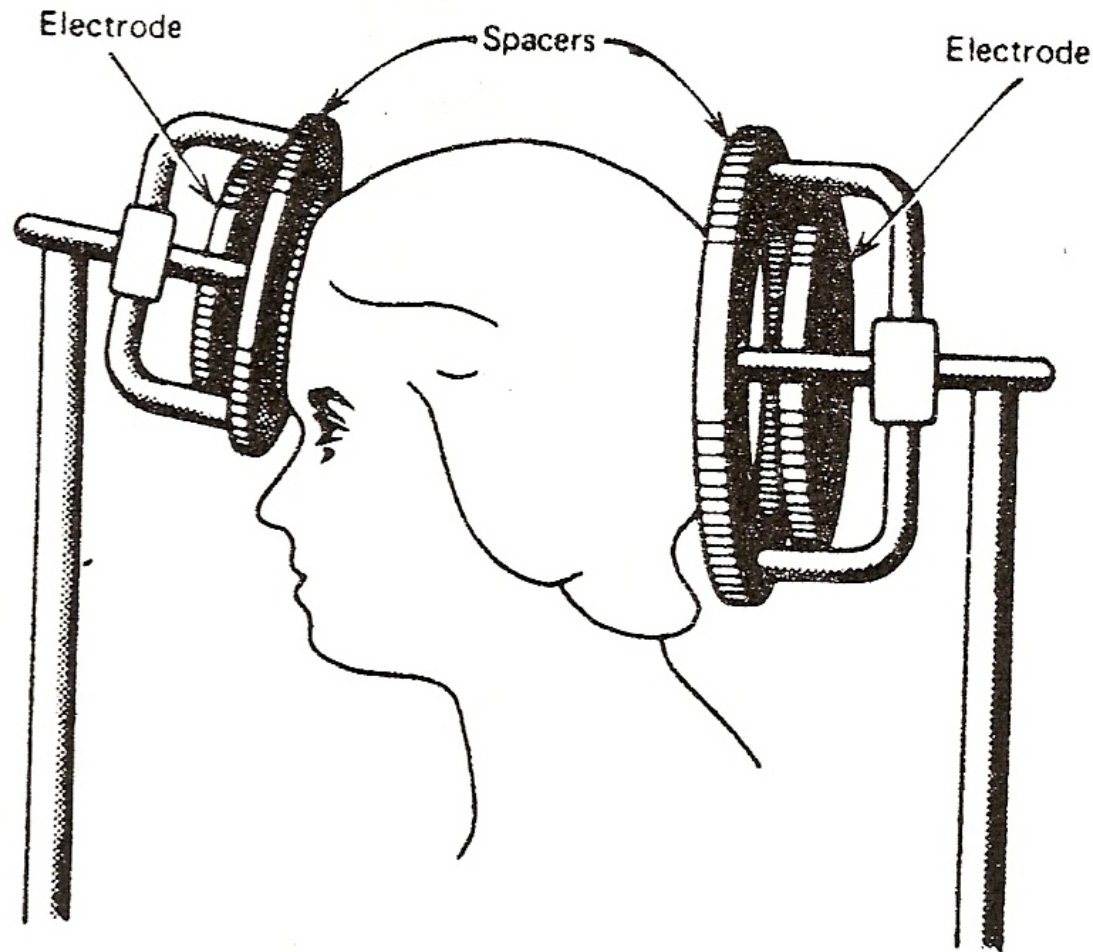


Figure 4.9. Location of capacitor plates for short wave diathermy. (Adapted from Lehmann, J.F.: "Diathermy," in Krusen, F.H., Kottke, F.J., and Ellwood, P.M. (Eds.): *Handbook of Physical Medicine and Rehabilitation*, 2nd edition. Philadelphia, W.B. Saunders Company, 1971, p. 286.)

B- By transferring short wave energy into the body by magnetic induction. This can be done either by placing a coil around the region to be treated Fig. (4.10) or by (pancake) coil placed near the part of the body to be treated Fig. (16.5). The alternating current in the coil produces an alternating magnetic field, consequently an alternating eddy currents are induced in the tissue and producing (Joule-heating).

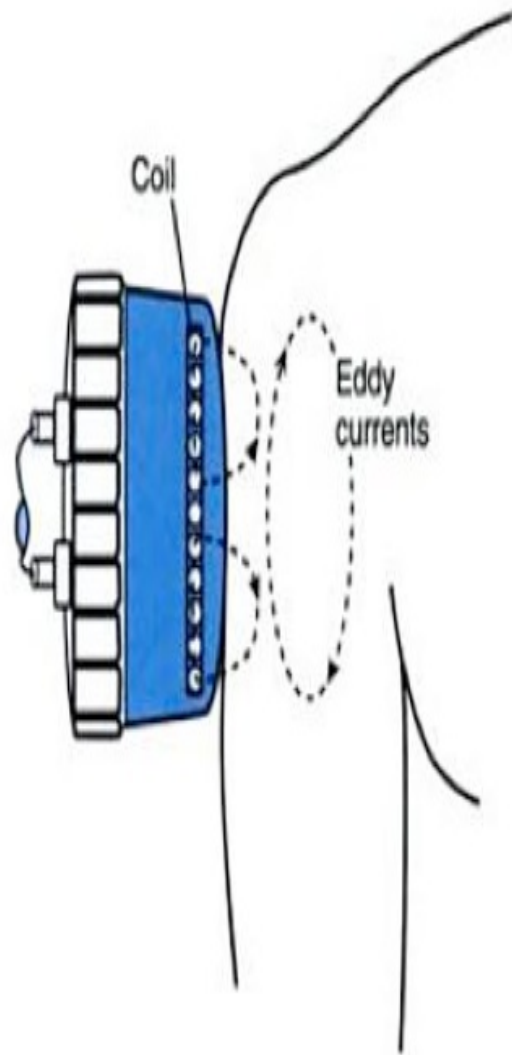


FIGURE 16.5. The magnetic field induced by short wave diathermy creates small eddy currents in the body tissues, resulting in heat.

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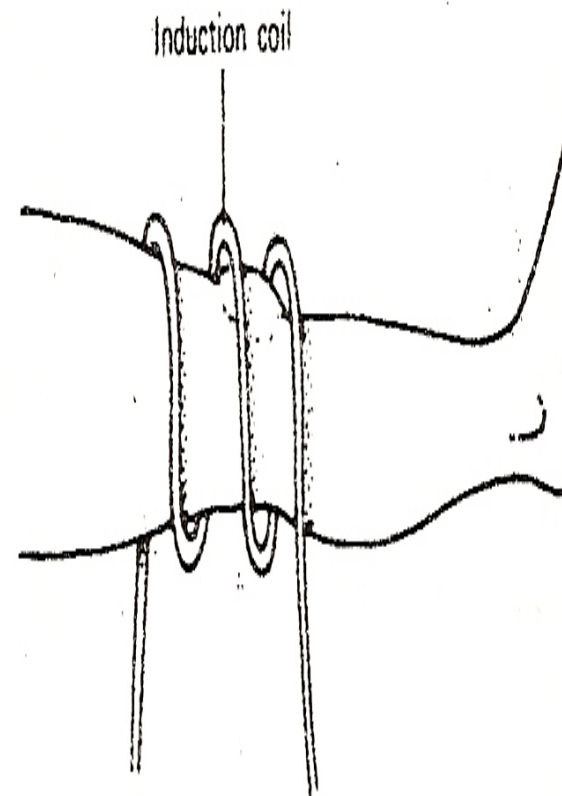


Figure 4.10. Location of induction coil around knee for short wave diathermy.

Short wave diathermy can penetrate deep into tissue. It can be used in relieving muscle spasms, protruded intervertebra discs pain, joints with minimal soft tissue coverage such as knee, elbow.

C-Microwave diathermy:

The microwave diathermy can be produced in special tube called (magnetron) and emitted from the applicator (antenna) which can be placed several inches from the region to be treated. Microwave can penetrate deep into the tissue causing heating. It is used in fractures, sprains, strains, injuries to tendons. The frequency used is 900MHz, which is found more effective than other frequencies in therapy. It causes more uniform heating around bony regions.

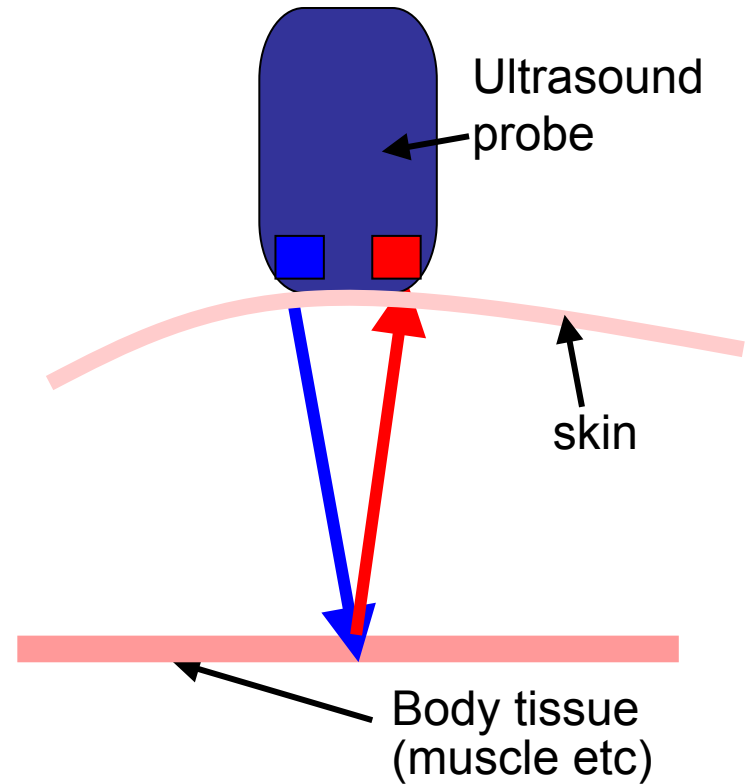
4-Ultrasonic waves:

These waves are different from **electromagnetic waves**. It produces mechanical vibrations inside tissue. It is the same as the sound waves but it has much higher frequencies (about 1MHz) with power of several watts per centimeter. It can move the tissue particles backward and forward with high frequency, in doing so it can increase the kinetic energy consequently it heats the tissue.

Ultrasound can be produced by **special transducers** placed in direct contact with the skin. It is used for relieving tightness and scarring occurring in joint diseases. It can dispose more heat in bones, as bones are better absorber for ultrasonic energy than soft tissue. It is also used in deep therapy.

Pulses of ultrasound are transmitted into the body by placing the vibrating crystal in close contact with the skin using water or a jelly paste to eliminate the air.

This gives a good coupling at the skin and greatly increases the transmission of the ultrasound into the body and of the echoes back to the detector.



1. What are the two primary therapeutic effects of heat therapy?
2. Which method of heat therapy involves direct contact to transfer heat?
3. What is the primary advantage of radiant heat therapy over conductive heating?
4. What frequency is commonly used in microwave diathermy?
5. Which type of diathermy uses electromagnetic waves?
6. How do ultrasonic waves generate heat in tissues?
7. What can excessive exposure to infrared radiation cause?
8. What is the purpose of the water or jelly paste when using ultrasound therapy?
9. Which type of heat therapy is most effective for relieving muscle spasms?
10. When is it generally recommended to use heat therapy?